

What are categories?

Topoi: The Categorical Analysis of Logic

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Goals

Walk away with

I.) Why categories matter

II.) Category Definition

III.) Set theory and Category Theory

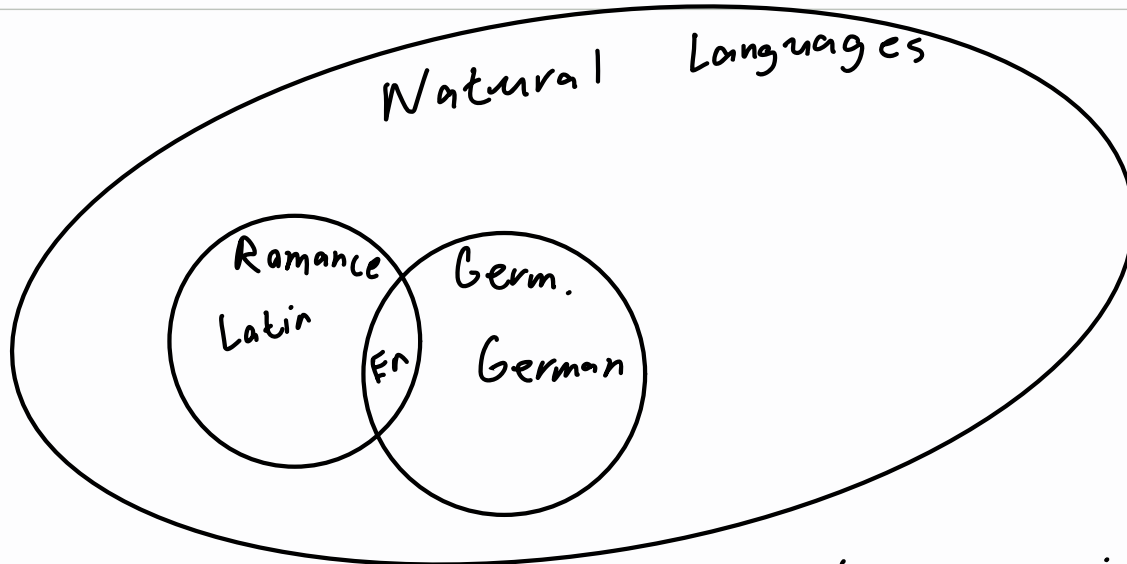
Why categories matter

Unification

Categories are general and apply to all mathematical structures.

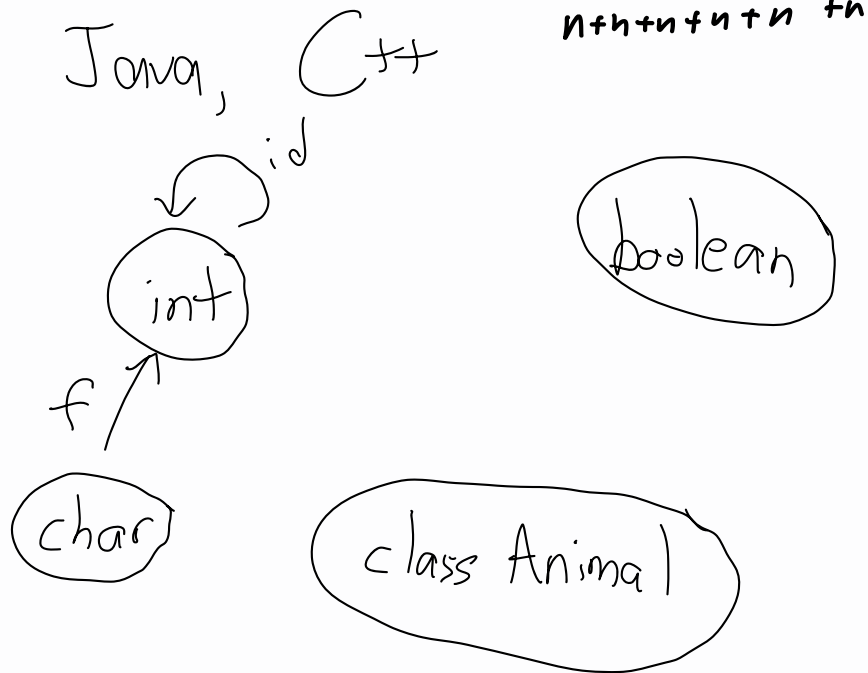
NOT Universal language

Comparisons with linguistics



- Disclaimer: Linguistics is describing existing languages, C.T. is for research!

Comparisons with programming

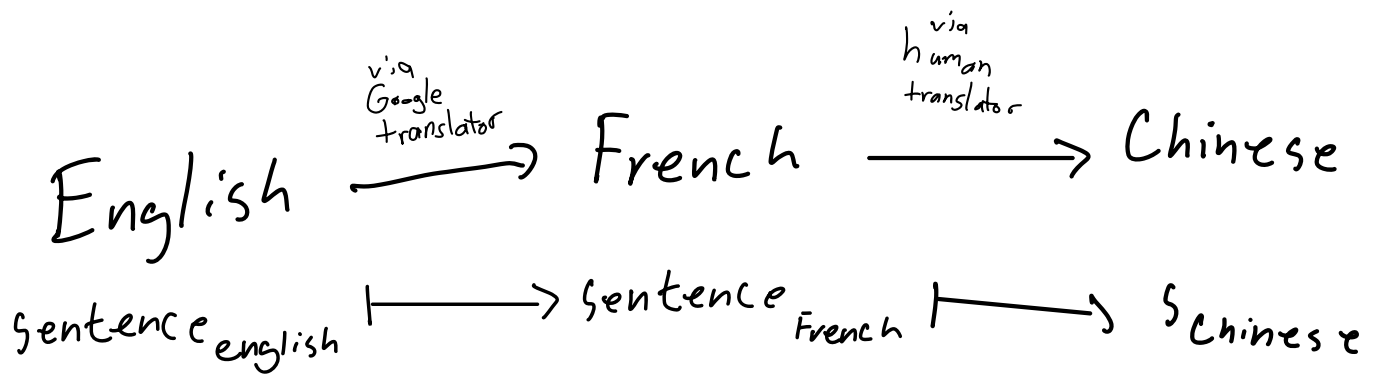


```
int comp(char c) {  
    return id(f(c));  
}
```

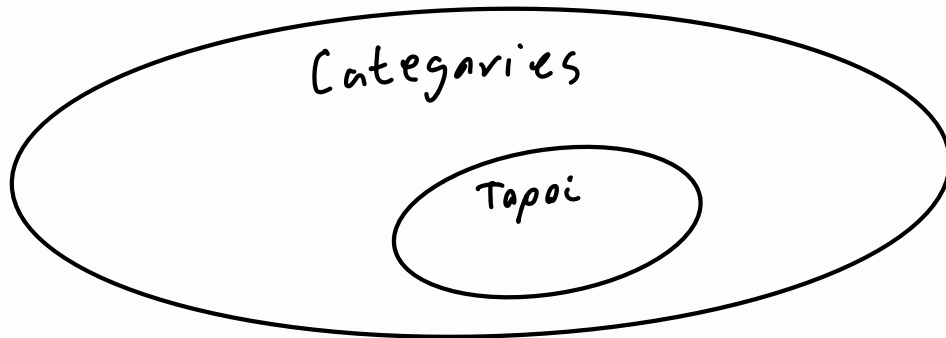
```
int f(char c) {  
    if (c == 'a')  
        return 42;  
    return 120;  
}
```

```
int id(int x) {  
    return x;  
}
```

"id ∘ f"



Talk about a lot of things at once



Usefulness

topos

- finite limits
- subobject classifier
- ...



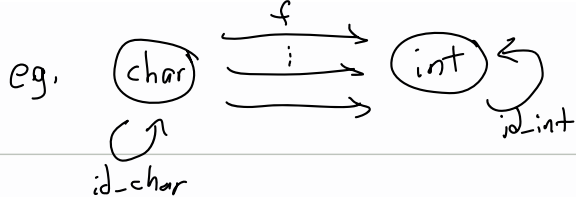
- power objects

Universal Properties

Categories in practice (informal examples)

Category definition and examples

Category definition



```

int id_int(int x) {
    return x;
}
  
```

```

char id_char(char c) {
    return c;
}
  
```

A category $\mathcal{C}$ has:

(01) $\text{Obj}(\mathcal{C})$ a collection of objects (0-cells)

(02) $\text{Mor}(\mathcal{C})$ a collection of morphisms (arrows, 1-cells, homomorphisms)

(03) $\text{dom}: \text{Mor}(\mathcal{C}) \rightarrow \text{Obj}(\mathcal{C})$

(04) $\text{cod}: \text{Mor}(\mathcal{C}) \rightarrow \text{Obj}(\mathcal{C})$

(05) Given $A \xrightarrow{f} B \xrightarrow{g} C$, we can compose to $A \xrightarrow{g \circ f} C$

(06) $\text{id}: \text{Obj}(\mathcal{C}) \rightarrow \text{Mor}(\mathcal{C})$

Category operations

(O3) each $\mathcal{C}$ -arrow f has a “domain” $\mathcal{C}$ -object $\text{dom } f$

$$\text{dom} : \text{Mor}(\mathcal{C}) \rightarrow \text{Obj}(\mathcal{C})$$

(O4) each $\mathcal{C}$ -arrow f has a “codomain” $\mathcal{C}$ -object $\text{cod } f$

$$\text{cod} : \text{Mor}(\mathcal{C}) \rightarrow \text{Obj}(\mathcal{C})$$

(O5) each path of $\mathcal{C}$ -arrows $A \xrightarrow{f} B \xrightarrow{g} C$ admits a “composite” $\mathcal{C}$ -arrow $g \circ f$

$$\circ : \text{Mor}(\mathcal{C})_{\text{dom} \times_{\text{cod}} \text{Mor}(\mathcal{C})} \rightarrow \text{Mor}(\mathcal{C})$$

(O6) each $\mathcal{C}$ -object X has an “identity” $\mathcal{C}$ -arrow 1_X

$$1 : \text{Obj}(\mathcal{C}) \rightarrow \text{Mor}(\mathcal{C})$$

Category axioms

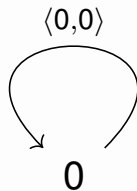
$$\mathcal{C} = (\text{Obj}(\mathcal{C}), \text{Mor}(\mathcal{C}), \text{dom}, \text{cod}, \circ, 1)$$

Label	Name	Axiom
(A1)	Domain of an identity:	$\text{dom } 1_X = X$
(A2)	Codomain of an identity:	$\text{cod } 1_X = X$
(A3)	Domain of a composite:	$\text{dom}(g \circ f) = \text{dom } f \iff \text{dom } g = \text{cod } f$
(A4)	Codomain of a composite:	$\text{cod}(g \circ f) = \text{cod } g$
(A5)	Right identity law:	$g \circ 1_{\text{dom } g} = g$
(A6)	Left identity law:	$1_{\text{cod } f} \circ f = f$
(A7)	Associativity law:	$h \circ (g \circ f) = (h \circ g) \circ f$ $\text{cod } f = \text{dom } g, \text{ cod } g = \text{dom } h \nearrow$

One-object Pre-ordered Category 1

$$\text{Obj}(1) = \{0\}$$

$$\text{Mor}(1) = \{\langle 0, 0 \rangle\}$$



$$\text{dom}(\langle 0, 0 \rangle) = 0 = \text{cod}(\langle 0, 0 \rangle)$$

$$\text{id}(0) = \langle 0, 0 \rangle$$

$$\begin{aligned} & \text{Mor}(1) \times_{\text{dom, cod}} \text{Mor}(1) \\ &= \{ \langle \langle 0, 0 \rangle, \langle 0, 0 \rangle \rangle \} \\ & \quad \downarrow 0 \\ & \{ \langle 0, 0 \rangle \} \end{aligned}$$

(A2) Does $\text{cod}(\text{id}(x)) = x$ for all $x \in \text{Obj}(1)$?

$$\text{cod}(\text{id}(0)) = \text{cod}(\langle 0, 0 \rangle) = 0$$



Pre-ordered Category 4

$$\cancel{\text{comp}(\langle p, q \rangle, \langle q, r \rangle)}$$

 or

$$\text{comp}(\langle q, r \rangle, \langle p, q \rangle)$$

$$p=0, q=1, r=2$$

$$\langle 0, 2 \rangle = \langle 1, 2 \rangle \circ \langle 0, 1 \rangle$$

$$\langle 0, 2 \rangle = \langle 0, 1 \rangle ; \langle 1, 2 \rangle$$

$$\text{ob}(4) = \{0, 1, 2, 3\}$$

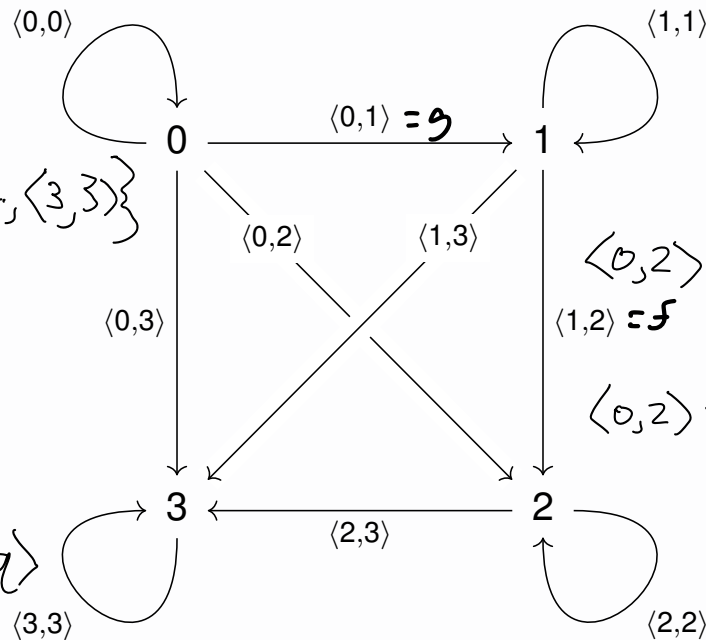
$$\text{Mor}(4) = \{\langle 0, 0 \rangle, \langle 0, 1 \rangle, \dots, \langle 3, 3 \rangle\}$$

$$\text{dom}(\langle m, n \rangle) = m$$

$$\text{cod}(\langle m, n \rangle) = n$$

$$\text{id}(n) = \langle n, n \rangle$$

$$\text{comp}(\underbrace{\langle p, q \rangle, \langle r, s \rangle}_{p=s}) = \langle r, q \rangle$$



One-object Category: $(\mathbb{N}, +, 0) \simeq \mathcal{C}$

$$\text{Obj}(\mathcal{C}) = \{\mathbb{N}\}$$

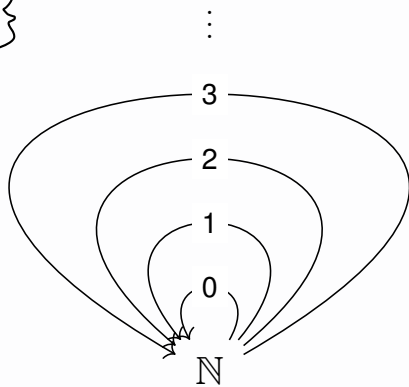
$$\text{Mor}(\mathcal{C}) = \mathbb{N} = \{0, 1, 2, 3, \dots\}$$

$$\text{dom}(n) = \mathbb{N} = \text{cod}(n)$$

$$\text{id}(\mathbb{N}) = 0$$

$$\text{comp}(m, n) = m + n$$

$$\begin{aligned} \text{dom}(m) &= \mathbb{N} \\ \text{cod}(n) &= \mathbb{N} \end{aligned}$$



laws of
arithmetic



this is a
valid category

Category $\text{Matr}(\mathbb{C})$

one object per natural number (dimensions of matrix)
morphisms are matrices

$$\text{Ob}(\text{Matr}(\mathbb{C})) = \mathbb{N}$$

$$\text{Mor}(\text{Matr}(\mathbb{C})) \ni$$

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_A, \underbrace{\begin{bmatrix} i & i \\ -1 & 3 \\ 2 & 2+i \end{bmatrix}}_B, \underbrace{[1 \ 2 \ 5 \ 6]}_C,$$

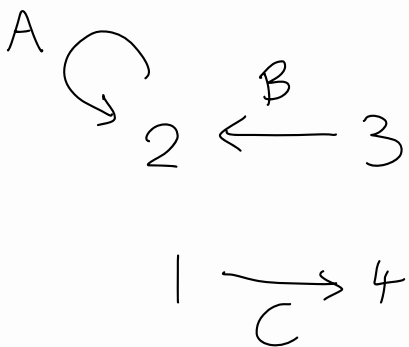
$$\underbrace{\begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}}_{n \times 0 \text{ matrix}}_D$$

$$\text{dom}(A) = \# \text{ of rows}$$

$$\text{cod}(A) = \# \text{ of cols}$$

$$A \circ B = AB$$

$$\text{id}(n) = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}_{n \times n}$$



Pre-ordered Category $(\mathcal{P}(X), \subseteq)$

Pre-ordered Category (Ordinals, $\leq$)

Sets and functions

Set

$$\text{Obj}(\text{Set}) = \text{Sets} \quad f = \langle A, B, R \rangle \text{ where } R \subseteq A \times B$$

$$\text{Mor}(\text{Set}) = \text{functions} = \langle \widehat{\text{Dom}}, \widehat{\text{Cod}}, \widehat{f} \rangle$$

$$\text{Dom}(f) = A$$

$$\text{Cod}(f) = B$$

$$\text{id}(\text{Set}) = x \mapsto x$$

$$x \mapsto x$$

$$\text{comp}(f \circ g) = f \circ g$$

Set definition

Sets in math vs. computer science

Why sets are special

Function definition

$$\lambda \in (0,1)$$

Relation

$$R \subseteq A \times B$$

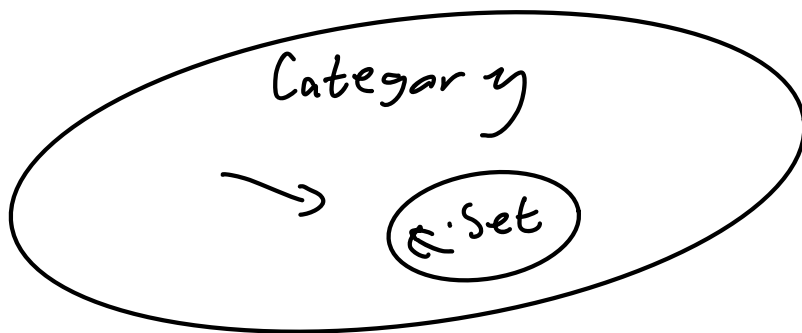
$$R := \{ (a, b) : a R b \}$$

Function

$$F \subseteq A \times B$$

$$F := \{ (a, b) : a F b \text{ and } a F c, a = c \}$$

Functions in math vs. computer science



Set-Theory

Set

Functions

injective

surjective

bijjective

Category

Object

Morphisms

monomorphisms = monic

epimorphisms = epic

isomorphisms = iso

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,

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,

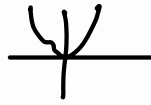
Relation definition

Sets and functions, categorically

Monomorphisms

$$f(x) = x^2$$

-1 and 1
1



injective function: $\forall x, y \in A$ if $f(x) = f(y)$
 $x = y$

monic

A morphism $f: a \rightarrow b$ s.t. $g, h: c \rightarrow a$

$$f \circ g = f \circ h$$

Epimorphisms

surjective
onto

A function $f: A \rightarrow B$ is surjective when

$$f(A) = B.$$

$$\forall b \in B, \exists a \in A \text{ s.t. } f(a) = b$$

$$\text{surjective} \Leftrightarrow (\forall b \in B)(\exists a \in A)(f(a) = b).$$

An epimorphism is right cancelable.

$$\text{i.e. } g \circ f = h \circ f$$

$$g, h: B \rightarrow C$$

implies

$$g = h.$$

Assumption: $g \circ f = h \circ f$.

Goal: $\forall x \in B$, does $g(x) = h(x)$?

given $x \in B$, write it as $x = f(a)$
for some $a \in A$. ★

$$\begin{aligned} g(x) &= g(f(a)) = (g \circ f)(a) = \\ &= (h \circ f)(a) \\ &= h(f(a)) \\ &= h(x) \end{aligned}$$

Isomorphisms

Bijjective if it is injective & surjective.

f bi $\Rightarrow f$ has an inverse.

bi $\Rightarrow$ iso

A morphism f is iso if there exists
some f^{-1} s.t.

$$f: A \rightarrow B$$

$$f^{-1}: B \rightarrow A$$

$$a \in A, \quad f^{-1}(f(a)) = id_A$$

$$f^{-1} \circ f = id_A$$

$$epic + monic \not\Rightarrow iso \quad f \circ f^{-1} = id_B$$

Isomorphism example
